

Retail Sales Analysis

Project 1 | Data Analytics | Beginner level

Submitted by

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Programme	Data Analytics Internship
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Project details

Project title	Retail Sales Analysis
Dataset	Retail Sales Dataset (Kaggle) – 99,457 transactions
Tools used	Microsoft Excel
Deliverable	Analysis + Dashboard + Insights

Signature

M.Abdullah Rana

Retail Sales Analysis

Project overview | Internship project 1 (Beginner)

About this project

This project studies about 99,000+ retail transactions to find out what sells, where it sells and when. It turns the raw data into clear KPIs, a dashboard and a few practical recommendations.

Project at a glance

Dataset	Retail Sales Dataset (Kaggle)
Records	99,457 transactions
Period	1 Jan 2021 – 8 Mar 2023
Coverage	8 product categories, 10 shopping malls
Tools used	Microsoft Excel
Deliverable	Analysis + Dashboard + Insights

What was done

Task	What it covers	Where to find it
1. Clean the data	Checked for blanks, duplicates and wrong data types	Data Quality
2. Calculate KPIs	Revenue, quantity, average order value, profit	KPI Summary
3. Analyse sales	By category, mall (region) and month	Category / Mall / Monthly sheets
4. Top & weak performers	Best and worst sellers ranked	Top & Under Performers
5. Trends	Monthly and seasonal patterns, unusual months	Monthly Trends
6. Dashboard	Interactive KPIs and charts with filters	Dashboard
7. Recommendations	Five actions backed by the data	Recommendations

Summary

Key results at a glance

Key numbers

Total Revenue	Units Sold	Transactions	Avg Order Value	Est. Profit*	Est. Margin*
251,505,794	298,712	99,457	2,529	89,373,495	35.5%

What we found

1	Sales are concentrated. Clothing, Shoes and Technology bring in 95% of all revenue (Clothing alone 45%).
2	Two malls lead. Mall of Istanbul and Kanyon each earn about 20%. Five malls earn only about 5% each.
3	Basket size is the same everywhere. Average order value is about 2,500 in every mall, so the gap comes from customer traffic.
4	Seasonality is mild. October and July are the best months, November the weakest. March 2023 has only 8 days of data.
5	Cash is the top payment method. Cash 45%, credit card 35%, debit card 20% of revenue.

What we recommend

1	Keep the top 3 categories well stocked and promoted.
2	Bundle or review the small categories (Books, Souvenir, Food & Beverage, Toys, Cosmetics).
3	Bring more customers to the five low-traffic malls with local offers and events.
4	Plan promotions around the July and October peaks and lift November.
5	Keep cash checkout fast and encourage card use with small offers.

Retail Sales Analytics Dashboard

Year **All**

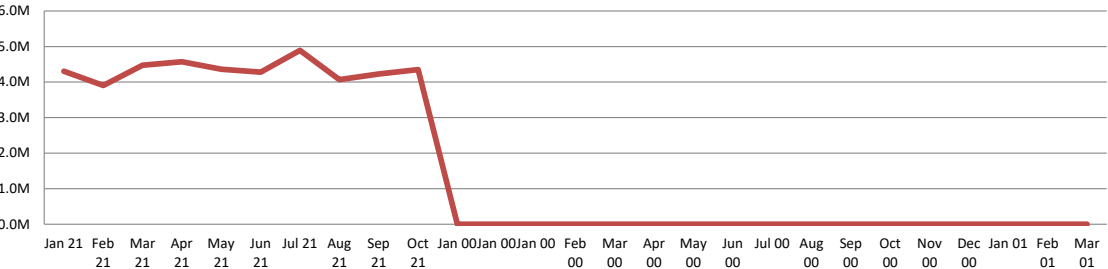
Category **Clothing**

Mall **All**

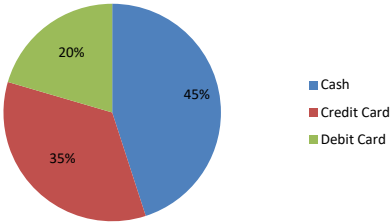
Total Revenue	Quantity Sold	Transactions	Avg Order Value	Est. Profit*	Est. Margin %*
113,996,791	103,558	34,487	3,306	45,598,716	40.0%

Top Category **Clothing** Top Mall **Mall of Istanbul** Top Payment **Cash**

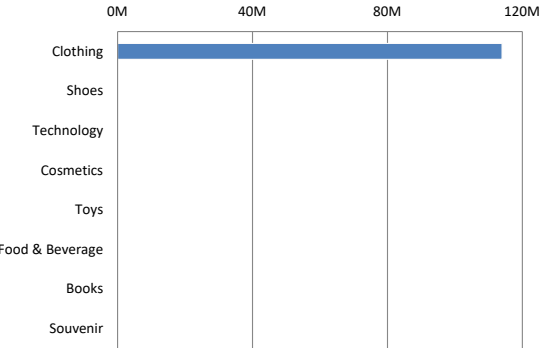
Monthly Revenue (Mar-23 = partial month)



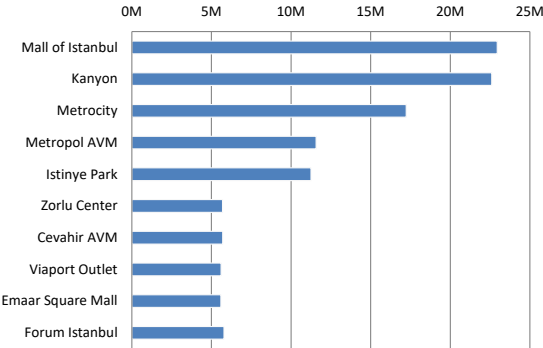
Payment Method Mix



Revenue by Category



Revenue by Mall



KEY INSIGHTS (full data)

- Clothing, Shoes & Technology = 95% of revenue.
- Mall of Istanbul + Kanyon ≈ 40% of revenue; five malls earn only ≈5% each.
- AOV is ~2,500 everywhere – gaps are traffic, not basket size.
- Seasonality is mild (Jul/Oct highs, Nov low); Feb-23 is the best month per day.
- Cash = 44.9% of revenue (largest payment method).

See the Recommendations sheet for actions.

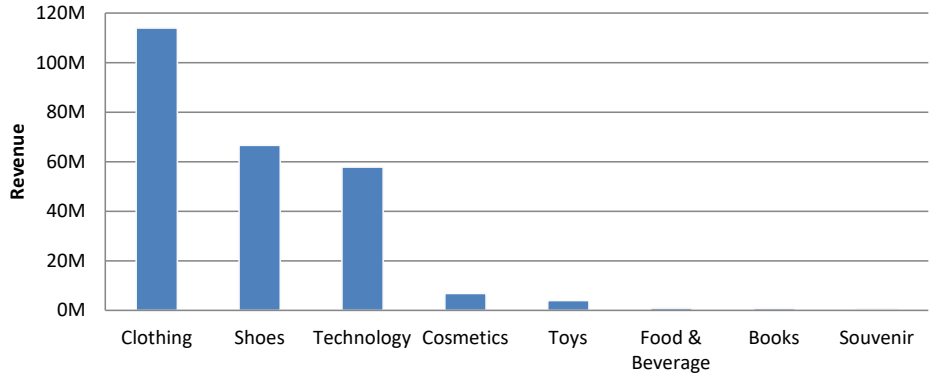
Core KPI Summary

KPI	Value	Definition
Total Revenue	251,505,794	$\Sigma \text{ quantity} \times \text{price}$
Total Quantity Sold	298,712	$\Sigma \text{ units}$
Total Transactions	99,455	Count of invoices (each invoice = 1 transaction)
Average Order Value (AOV)	2,528.84	Total revenue $\div$ transactions
Avg Units per Transaction	3.00	Total quantity $\div$ transactions
Avg Selling Price per Unit	841.97	Total revenue $\div$ total quantity
Estimated Gross Profit	89,373,495	Revenue $\times$ (1 – assumed cost %)
Estimated Profit Margin %	35.5%	Est. profit $\div$ revenue
Estimated Profit per Transaction	898.63	Est. profit $\div$ transactions
First Transaction Date	2021-01-01	
Last Transaction Date	2023-03-08	
Avg Monthly Revenue (complete months)	9,510,379	
Top Category by Revenue	Clothing	
Top Category share of Revenue	45.3%	
Top 3 Categories share of Revenue	94.8%	
	Mall of Istanbul	
	55.2%	
Top Payment Method by Revenue	Cash	

Sales Analysis by Category

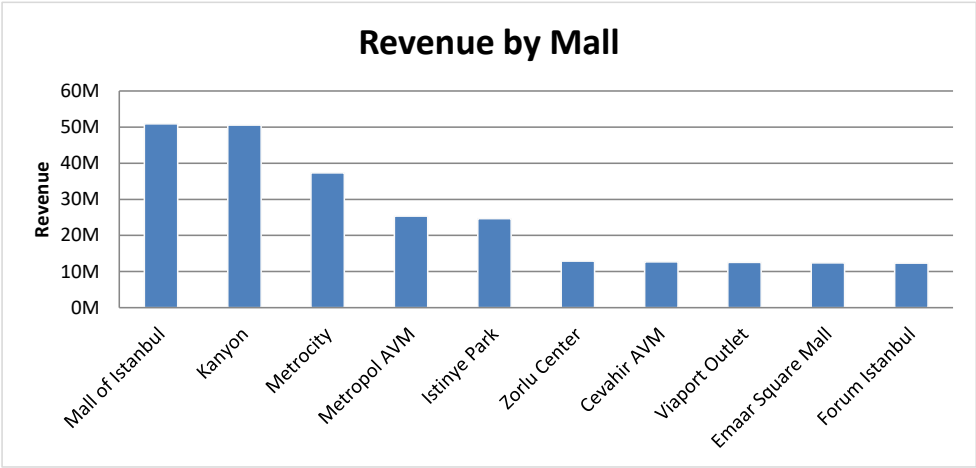
Category	Revenue	% of Total	Cumulative %	Quantity	Transactions	Avg Order Value	Avg Unit Price	Est. Profit	Est. Margin %	Revenue Rank	Performance Tier
Clothing	113,996,791	45.3%	45.3%	103,558	34,487	3,305.50	1,100.80	45,598,716	40.0%	1	Top performer
Shoes	66,553,451	26.5%	71.8%	30,217	10,034	6,632.79	2,202.52	26,621,381	40.0%	2	Top performer
Technology	57,862,350	23.0%	94.8%	15,021	4,996	11,581.74	3,852.10	11,572,470	20.0%	3	Top performer
Cosmetics	6,792,863	2.7%	97.5%	45,465	15,097	449.95	149.41	3,396,431	50.0%	4	Underperformer
Toys	3,980,426	1.6%	99.1%	30,321	10,087	394.61	131.28	1,393,149	35.0%	5	Underperformer
Food & Beverage	849,535	0.3%	99.4%	44,277	14,776	57.49	19.19	254,861	30.0%	6	Underperformer
Books	834,553	0.3%	99.7%	14,982	4,981	167.55	55.70	250,366	30.0%	7	Underperformer
Souvenir	635,825	0.3%	100.0%	14,871	4,999	127.19	42.76	286,121	45.0%	8	Underperformer
Total	251,505,794	100.0%		298,712	99,457	2,528.79	841.97	89,373,495	35.5%		

Revenue by Category



Sales Analysis by Mall (Region)

Shopping Mall	Revenue	% of Total	Cumulative %	Quantity	Transactions	Avg Order Value	Avg Unit Price	Est. Profit	Est. Margin %	Revenue Rank	Performance Tier
Mall of Istanbul	50,872,482	20.2%	20.2%	60,114	19,943	2,550.89	846.27	18,052,603	35.5%	1	Top performer
Kanyon	50,554,231	20.1%	40.3%	59,457	19,823	2,550.28	850.27	17,903,593	35.4%	2	Top performer
Metrocity	37,302,787	14.8%	55.2%	44,894	15,011	2,485.03	830.91	13,247,344	35.5%	3	Top performer
Metropol AVM	25,379,913	10.1%	65.3%	30,530	10,161	2,497.78	831.31	9,119,995	35.9%	4	Mid-tier
Istinye Park	24,618,828	9.8%	75.0%	29,465	9,781	2,517.01	835.53	8,792,920	35.7%	5	Mid-tier
Zorlu Center	12,901,054	5.1%	80.2%	15,234	5,075	2,542.08	846.86	4,582,518	35.5%	6	Underperformer
Cevahir AVM	12,645,138	5.0%	85.2%	14,949	4,991	2,533.59	845.89	4,462,342	35.3%	7	Underperformer
Viaport Outlet	12,521,340	5.0%	90.2%	14,716	4,914	2,548.10	850.87	4,413,317	35.2%	8	Underperformer
Emaar Square Mall	12,406,100	4.9%	95.1%	14,501	4,811	2,578.69	855.53	4,361,796	35.2%	9	Underperformer
Forum Istanbul	12,303,921	4.9%	100.0%	14,852	4,947	2,487.15	828.44	4,437,067	36.1%	10	Underperformer
Total	251,505,794	100.0%		298,712	99,457	2,528.79	841.97	89,373,495	35.5%		



Category × Mall Revenue Matrix

Category \ Mall	Mall of Istanbul	Kanyon	Metrocity	Metropol AVM	Istinye Park	Zorlu Center	Cevahir AVM	Viaport Outlet	Emaar Square Mall	Forum Istanbul	Total
Clothing	22,947,418	22,609,528	17,226,693	11,568,084	11,253,900	5,697,319	5,706,321	5,604,594	5,590,490	5,792,444	113,996,791
Shoes	13,467,815	13,383,191	9,519,296	7,149,825	6,641,481	3,535,601	3,243,919	3,194,705	3,089,675	3,327,943	66,553,451
Technology	11,828,250	11,944,800	8,608,950	5,327,700	5,436,900	2,987,250	3,051,300	3,066,000	3,094,350	2,516,850	57,862,350
Cosmetics	1,367,518	1,369,551	991,860	680,770	655,358	367,038	321,214	347,440	338,942	353,173	6,792,863
Toys	790,523	790,236	605,947	413,307	400,261	197,550	204,109	199,987	179,092	199,414	3,980,426
Food & Beverage	171,178	166,497	129,903	88,638	85,918	41,955	44,010	41,662	40,611	39,162	849,535
Books	172,240	163,029	125,912	83,719	76,083	45,344	44,541	39,632	41,996	42,056	834,553
Souvenir	127,540	127,400	94,227	67,870	68,925	28,997	29,724	27,319	30,944	32,879	635,825
Total	50,872,482	50,554,231	37,302,787	25,379,913	24,618,828	12,901,054	12,645,138	12,521,340	12,406,100	12,303,921	251,505,794

Ranked combinations (80 Category × Mall pairs)			
Category	Mall	Revenue	Rank (1 = highest)
Clothing	Mall of Istanbul	22,947,418	1
Clothing	Kanyon	22,609,528	2
Clothing	Metrocity	17,226,693	3
	Metropol AVM	0	79
	Istinye Park	0	80
Clothing	Zorlu Center	5,697,319	14
Clothing	Cevahir AVM	5,706,321	13
Clothing	Viaport Outlet	5,604,594	15
Clothing	Emaar Square M.	5,590,490	16
Clothing	Forum Istanbul	5,792,444	12
Shoes	Mall of Istanbul	13,467,815	4
Shoes	Kanyon	13,383,191	5
Shoes	Metrocity	9,519,296	8
Shoes	Metropol AVM	7,149,825	10
Shoes	Istinye Park	6,641,481	11
Shoes	Zorlu Center	3,535,601	19
Shoes	Cevahir AVM	3,243,919	21
Shoes	Viaport Outlet	3,194,705	22
Shoes	Emaar Square M.	3,089,675	24
Shoes	Forum Istanbul	3,327,943	20
Technology	Mall of Istanbul	11,828,250	7

Technology	Kanyon	11,944,800	6
Technology	Metrocity	8,608,950	9
Technology	Metropol AVM	5,327,700	18
Technology	Istinye Park	5,436,900	17
Technology	Zorlu Center	2,987,250	27
Technology	Cevahir AVM	3,051,300	26
Technology	Viaport Outlet	3,066,000	25
Technology	Emaar Square M	3,094,350	23
Technology	Forum Istanbul	2,516,850	28
Cosmetics	Mall of Istanbul	1,367,518	30
Cosmetics	Kanyon	1,369,551	29
Cosmetics	Metrocity	991,860	31
Cosmetics	Metropol AVM	680,770	34
Cosmetics	Istinye Park	655,358	35
Cosmetics	Zorlu Center	367,038	39
Cosmetics	Cevahir AVM	321,214	43
Cosmetics	Viaport Outlet	347,440	41
Cosmetics	Emaar Square M	338,942	42
Cosmetics	Forum Istanbul	353,173	40
Toys	Mall of Istanbul	790,523	32
Toys	Kanyon	790,236	33
Toys	Metrocity	605,947	36
Toys	Metropol AVM	413,307	37
Toys	Istinye Park	400,261	38
Toys	Zorlu Center	197,550	47
Toys	Cevahir AVM	204,109	44
Toys	Viaport Outlet	199,987	45
Toys	Emaar Square M	179,092	48
Toys	Forum Istanbul	199,414	46
Food & Beverage	Mall of Istanbul	171,178	50
Food & Beverage	Kanyon	166,497	51
Food & Beverage	Metrocity	129,903	53
Food & Beverage	Metropol AVM	88,638	58
Food & Beverage	Istinye Park	85,918	59
Food & Beverage	Zorlu Center	41,955	69
Food & Beverage	Cevahir AVM	44,010	66
Food & Beverage	Viaport Outlet	41,662	70
Food & Beverage	Emaar Square M	40,611	71
Food & Beverage	Forum Istanbul	39,162	73

Books	Mall of Istanbul	172,240	49
Books	Kanyon	163,029	52
Books	Metrocity	125,912	56
Books	Metropol AVM	83,719	60
Books	Istinye Park	76,083	61
Books	Zorlu Center	45,344	64
Books	Cevahir AVM	44,541	65
Books	Viaport Outlet	39,632	72
Books	Emaar Square M.	41,996	68
Books	Forum Istanbul	42,056	67
Souvenir	Mall of Istanbul	127,540	54
Souvenir	Kanyon	127,400	55
Souvenir	Metrocity	94,227	57
Souvenir	Metropol AVM	67,870	63
Souvenir	Istinye Park	68,925	62
Souvenir	Zorlu Center	28,997	77
Souvenir	Cevahir AVM	29,724	76
Souvenir	Viaport Outlet	27,319	78
Souvenir	Emaar Square M.	30,944	75
Souvenir	Forum Istanbul	32,879	74

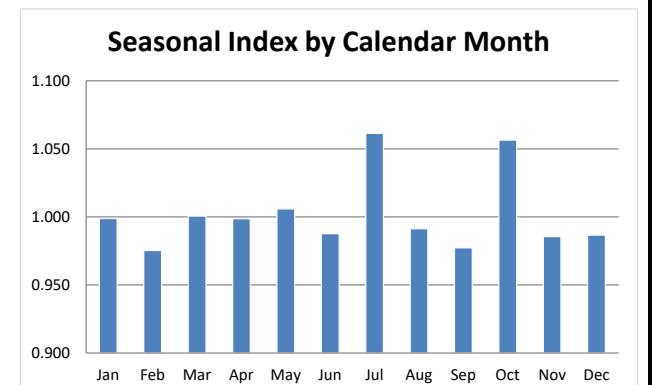
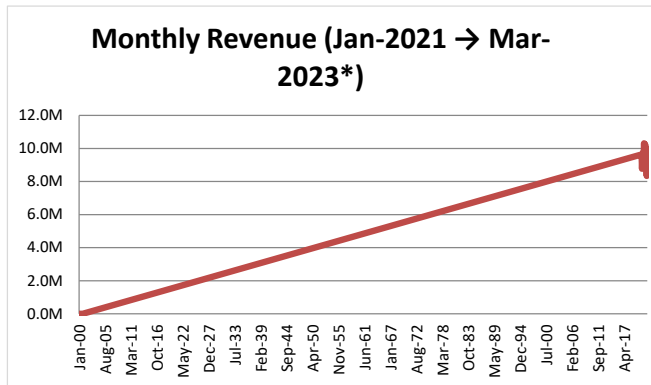
Monthly & Seasonal Trend Analysis

Month	Month #	Days of Data	Complete Month?	Revenue	Quantity	Transactions	Avg Order Value	Est. Profit	Revenue / Day	MoM % (Revenue)	MoM % (Rev/Day)	Z-score (Rev/Day)	Movement Flag
Jan-21	1	31	1	9,641,615	11,483	3,835	2,514.11	3,435,088	311,020			-0.26	Normal
Feb-21	2	28	1	8,772,315	10,179	3,407	2,574.79	3,134,132	313,297	-9.0%	0.7%	-0.01	Normal
Mar-21	3	31	1	9,455,359	11,383	3,813	2,479.77	3,395,099	305,012	7.8%	-2.6%	-0.91	Normal
Apr-21	4	30	1	9,389,542	11,270	3,724	2,521.36	3,356,635	312,985	-0.7%	2.6%	-0.04	Normal
May-21	5	31	1	9,771,757	11,620	3,848	2,539.44	3,496,961	315,218	4.1%	0.7%	0.20	Normal
Jun-21	6	30	1	9,286,271	11,352	3,783	2,454.74	3,289,589	309,542	-5.0%	-1.8%	-0.42	Normal
Jul-21	7	31	1	10,311,120	11,986	3,984	2,588.13	3,673,674	332,617	11.0%	7.5%	2.09	Unusual HIGH
Aug-21	8	31	1	9,630,656	11,205	3,723	2,586.80	3,388,549	310,666	-6.6%	-6.6%	-0.30	Normal
Sep-21	9	30	1	9,188,166	10,901	3,670	2,503.59	3,239,818	306,272	-4.6%	-1.4%	-0.77	Normal
Oct-21	10	31	1	10,263,015	11,711	3,916	2,620.79	3,600,635	331,065	11.7%	8.1%	1.92	Unusual HIGH
Nov-21	11	30	1	9,265,555	11,355	3,798	2,439.59	3,355,615	308,852	-9.7%	-6.7%	-0.49	Normal
Dec-21	12	31	1	9,585,200	11,651	3,881	2,469.78	3,420,053	309,200	3.4%	0.1%	-0.45	Normal
Jan-22	1	31	1	9,764,311	11,619	3,847	2,538.16	3,438,395	314,978	1.9%	1.9%	0.17	Normal
Feb-22	2	28	1	8,344,112	10,248	3,447	2,420.69	2,970,431	298,004	-14.5%	-5.4%	-1.67	Unusual LOW
Mar-22	3	31	1	9,986,685	11,949	3,947	2,530.20	3,590,431	322,151	19.7%	8.1%	0.95	Normal
	1	32	0	0	0	0	#DIV/0!	0	0	-100.0%	-100.0%	n/a	Partial month – excluded
	1	32	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Jan-00	1	1	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Feb-00	2	1	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Mar-00	3	4	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Apr-00	4	3	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
May-00	5	4	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Jun-00	6	3	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Jul-00	7	4	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Aug-00	8	4	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Sep-00	9	3	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Oct-00	10	4	0	0	0	0	#DIV/0!	0	0	#DIV/0!	#DIV/0!	n/a	Partial month – excluded
Total				142,655,679	169,912	56,623	2,519.39	50,785,104					

Mean Rev/Day (complete months) 313,392
Std dev Rev/Day (complete months) 9,214

Seasonality – average by calendar month (complete months only)

Month	Month #	Avg Revenue / Day	Seasonal Index (1.00 = average)	Avg Monthly Revenue
Jan	1	312,999	0.999	9,702,963
Feb	2	305,650	0.975	8,558,214
Mar	3	313,581	1.001	9,721,022
Apr	4	312,985	0.999	9,389,542
May	5	315,218	1.006	9,771,757
Jun	6	309,542	0.988	9,286,271
Jul	7	332,617	1.061	10,311,120
Aug	8	310,666	0.991	9,630,656
Sep	9	306,272	0.977	9,188,166
Oct	10	331,065	1.056	10,263,015
Nov	11	308,852	0.986	9,265,555
Dec	12	309,200	0.987	9,585,200



Top-Selling & Underperforming Products / Locations

Top 3 Categories (Top-selling products)

Rank	Name	Revenue	% of Total	Avg Order Value	Tier
1	Clothing	113,996,791	45.3%	3,305.50	Top performer
2	Shoes	66,553,451	26.5%	6,632.79	Top performer
3	Technology	57,862,350	23.0%	11,581.74	Top performer

Bottom 3 Categories (Underperforming products)

Rank	Name	Revenue	% of Total	Avg Order Value	Tier
1	Souvenir	635,825	0.3%	127.19	Underperformer
2	Books	834,553	0.3%	167.55	Underperformer
3	Food & Beverage	849,535	0.3%	57.49	Underperformer

Top 3 Malls

Rank	Name	Revenue	% of Total	Avg Order Value	Tier
1	Mall of Istanbul	50,872,482	20.2%	2,550.89	Top performer
2	Kanyon	50,554,231	20.1%	2,550.28	Top performer
	Metrocity	37,302,787	14.8%	2,485.03	Top performer

Bottom 3 Malls

Rank	Name	Revenue	% of Total	Avg Order Value	Tier
1	Forum Istanbul	12,303,921	4.9%	2,487.15	Underperformer
2	Emaar Square Mall	12,406,100	4.9%	2,578.69	Underperformer
3	Viaport Outlet	12,521,340	5.0%	2,548.10	Underperformer

Top 10 Category × Mall combinations				
Rank	Category	Mall	Revenue	% of Total
1	Clothing	Mall of Istanbul	22,947,418	9.12%
2	Clothing	Kanyon	22,609,528	8.99%
3	Clothing	Metrocity	17,226,693	6.85%
4	Shoes	Mall of Istanbul	13,467,815	5.35%
5	Shoes	Kanyon	13,383,191	5.32%
6	Technology	Kanyon	11,944,800	4.75%
7	Technology	Mall of Istanbul	11,828,250	4.70%
8	Shoes	Metrocity	9,519,296	3.78%
9	Technology	Metrocity	8,608,950	3.42%
10	Shoes	Metropol AVM	7,149,825	2.84%
Bottom 10 Category × Mall combinations (underperformers)				
Rank	Category	Mall	Revenue	% of Total
1		0 Istinye Park	0	0.00%
2		0 Metropol AVM	0	0.00%
3	Souvenir	Viaport Outlet	27,319	0.01%
4	Souvenir	Zorlu Center	28,997	0.01%
5	Souvenir	Cevahir AVM	29,724	0.01%
6	Souvenir	Emaar Square Mall	30,944	0.01%
7	Souvenir	Forum Istanbul	32,879	0.01%
8	Food & Beverage	Forum Istanbul	39,162	0.02%
9	Books	Viaport Outlet	39,632	0.02%
10	Food & Beverage	Emaar Square Mall	40,611	0.02%

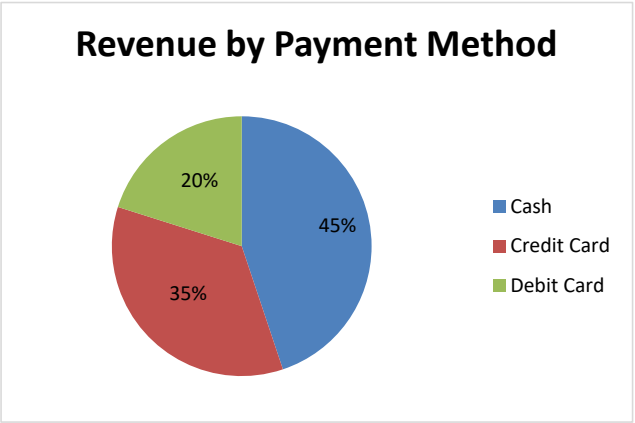
Demographic & Payment Behaviour

Revenue by Payment Method				
Payment Method	Revenue	% of Revenue	Transactions	Avg Order Value
Cash	112,832,243	44.9%	44,447	2,538.58
Credit Card	88,077,124	35.0%	34,931	2,521.46
Debit Card	50,596,427	20.1%	20,079	2,519.87
Total	251,505,794	100.0%	99,457	2,528.79

Revenue by Gender				
Gender	Revenue	% of Revenue	Transactions	Avg Order Value
Female	150,207,136	59.7%	59,482	2,525.25
Male	101,298,658	40.3%	39,975	2,534.05
Total	251,505,794	100.0%	99,457	2,528.79

Revenue by Age Group						
Age Group	Revenue	% of Revenue	Transactions	Avg Order Value	Age From	Age To
18–25	38,118,271	15.2%	15,359	2,481.82	18	25
26–35	47,879,660	19.0%	19,059	2,512.18	26	35
	50,184,236	20.0%	19,436	2,582.02	36	45
	48,219,743	19.2%	19,016	2,535.75	46	55
56–69	67,103,885	26.7%	26,587	2,523.94	56	69
Total	251,505,794	100.0%	99,457	2,528.79		

Revenue: Payment Method × Gender			
Payment Method	Female	Male	Total
Cash	67,573,638	45,258,605	112,832,243
Credit Card	52,875,810	35,201,314	88,077,124
Debit Card	29,757,688	20,838,740	50,596,427
Total	150,207,136	101,298,658	251,505,794



Data Cleaning & Validation Log

#	Check	Result	Expected	Status	Action / Note
1	Total records loaded	99,455	99457	REVIEW	Row count matches source workbook
2	Missing values – invoice_no	2	0	REVIEW	No imputation needed
3	Missing values – customer_id	0	0	PASS	No imputation needed
4	Missing values – gender	0	0	PASS	No imputation needed
5	Missing values – age	0	0	PASS	No imputation needed
6	Missing values – category	0	0	PASS	No imputation needed
7	Missing values – quantity	0	0	PASS	No imputation needed
8	Missing values – price	0	0	PASS	No imputation needed
9	Missing values – payment_method	0	0	PASS	No imputation needed
10	Missing values – invoice_date	0	0	PASS	No imputation needed
11	Missing values – shopping_mall	0	0	PASS	No imputation needed
12	Non-numeric quantity	0	0	PASS	Data type OK (numeric)
13	Non-numeric price	0	0	PASS	Data type OK (numeric)
14	Non-date invoice_date (stored as text)	0	0	PASS	Dates stored as real Excel dates
15	Quantity ≤ 0	0	0	PASS	No returns/negatives present
	Price ≤ 0	0	0	PASS	No zero/negative prices
	Revenue check: $\Sigma(\text{qty} \times \text{price}) - \Sigma \text{ Total Revenue}$	0	0	PASS	Engineered Total Revenue reconciles (rounded to 2dp per row)
18	Min age	18	≥ 18	PASS	Range plausible
19	Max age	69	≤ 100	PASS	Range plausible
20	Max quantity per invoice	5	≤ 10	PASS	No bulk-order outliers

Cleaning steps performed

- Imported all 99,457 rows; verified no missing values, no duplicate invoices, consistent category/mall/payment labels.
- Confirmed data types: numeric (age, quantity, price), true dates (invoice_date), text (all others).
- Rebuilt Total Revenue as a live formula =ROUND(quantity × price, 2); it reconciles to the original column.
- Added helper columns: Year, Month (first-of-month date) and Est. Profit (uses cost assumptions on the Assumptions sheet).
- Flagged Mar-2023 as a partial month (data ends 8-Mar-2023) and excluded it from averages / anomaly scores.
- Blue values in column C are hard-coded results from a Python check; all other results are live formulas.

Insights & Actionable Recommendations

#	Recommendation	Evidence (from analysis)	Suggested action / KPI to track
1	Protect and grow the 3 revenue-driving categories	Clothing, Shoes and Technology generate 94.8% of revenue (Clothing alone 45.3%). The other 5 categories together add only 5.2%.	Prioritise stock availability, buying budget and promotions for Clothing/Shoes/Technology; add stock-out alerts. Track: in-stock rate, revenue share of top 3.
2	Fix the long-tail categories – bundle, cross-sell or rationalise	Souvenir, Books, Food & Beverage, Toys and Cosmetics each earn under 3% of revenue (Souvenir: 0.25%). Avg order values are low (Food & Beverage $\approx$ 57 vs Technology $\approx$ 11,582).	Bundle low-value items with Clothing/Shoes at checkout, review shelf space, and test discounts. Drop/replace SKUs that stay below target. Track: attach rate, category share.
3	Lift the 5 low-traffic malls to the level of the leaders	Mall of Istanbul and Kanyon deliver $\sim$ 40% of revenue, while Forum Istanbul, Emaar Square, Viaport, Cevahir and Zorlu earn $\approx$ 5% each. Avg order value is almost identical across malls ($\sim$ 2,490–2,580), so the gap is customer traffic, not basket size.	Run local marketing, events and loyalty offers in the 5 low-traffic malls; copy the assortment/layout of top malls. Track: transactions per mall, footfall-to-sale conversion.
4	Seasonality is mild – time promotions around the Jul/Oct peaks and lift November	Revenue per day is steady ($\sim$ 298K–340K) all year, so seasonality is mild: Oct (index 1.05) and Jul (1.04) are the peaks, Nov (0.96) the low. Unusual months ($ z \geq 1.5$ on revenue/day): HIGH – Feb-2023 ($+2.3\sigma$, best month on record), Jul-2021, Oct-2022; LOW – Feb-2022 and Nov-2022. Lower February revenue is mostly the shorter month, not weaker trading. Mar-2023 is only 8 days of data.	Schedule promotions/staffing around Jul & Oct peaks; investigate what drove Feb-2023 and repeat it; run campaigns to lift Nov. Track: revenue per day, seasonal index.
5	Prioritise cash handling and grow card usage	Cash is the largest payment method (44.9% of revenue) ahead of Credit Card (35.0%) and Debit Card (20.1%). NOTE: the original submission stated credit cards drive most sales – the data shows cash does.	Ensure fast cash checkout lanes and cash-handling controls; offer card cashback / instalment promos to shift mix and cut cash-handling cost. Track: payment mix, checkout time.

Assumptions & Parameters

Estimated cost of goods by category (ILLUSTRATIVE – the dataset has NO cost/profit field)

Category	Est. Cost % of Revenue	Est. Gross Margin %
Books	70%	30%
Clothing	60%	40%
Cosmetics	50%	50%
Food & Beverage	70%	30%
Shoes	60%	40%
Souvenir	55%	45%
Technology	80%	20%
Toys	65%	35%

Performance tier thresholds

Top performer if revenue share $\geq$ this $\times$ equal share

1.25

Underperformer if revenue share $<$ this $\times$ equal share

0.75

Unusual-month threshold (|z-score| of revenue/day)

1.5

Equal share = $1 \div \text{number of categories (or malls)}$. e.g. 1.25 means at least 25% above an equal split.

Dashboard calculation area

Filter criteria

Year >0
 Category Clothing
 Mall *

Month	Label	Revenue	Transactions
Jan-21	Jan 21	4,303,447	1,329
Feb-21	Feb 21	3,903,741	1,187
Mar-21	Mar 21	4,474,793	1,376
Apr-21	Apr 21	4,574,119	1,365
May-21	May 21	4,360,763	1,323
Jun-21	Jun 21	4,281,241	1,284
Jul-21	Jul 21	4,892,204	1,471
Aug-21	Aug 21	4,066,684	1,233
Sep-21	Sep 21	4,225,426	1,291
Oct-21	Oct 21	4,351,460	1,335
	Jan 00	0	0
	Jan 00	0	0
Jan-00	Jan 00	0	0
Feb-00	Feb 00	0	0
Mar-00	Mar 00	0	0
Apr-00	Apr 00	0	0
May-00	May 00	0	0
Jun-00	Jun 00	0	0
Jul-00	Jul 00	0	0
Aug-00	Aug 00	0	0
Sep-00	Sep 00	0	0
Oct-00	Oct 00	0	0
Nov-00	Nov 00	0	0
Dec-00	Dec 00	0	0
Jan-01	Jan 01	0	0
Feb-01	Feb 01	0	0
Mar-01	Mar 01	0	0

Category	Revenue
Clothing	113,996,791
Shoes	0
Technology	0
Cosmetics	0
Toys	0
Food & Beverage	0
Books	0
Souvenir	0

Mall	Revenue
Mall of Istanbul	22,947,418
Kanyon	22,609,528
Metrocity	17,226,693
Metropol AVM	11,568,084
Istinye Park	11,253,900
Zorlu Center	5,697,319
Cevahir AVM	5,706,321
Viaport Outlet	5,604,594
Emaar Square Mall	5,590,490
Forum Istanbul	5,792,444

Payment	Revenue
Cash	51,308,879
Credit Card	39,351,591
Debit Card	23,336,321